

Test Cases

Bank Account Test Cases

Test ID	TC001
Test Name	Valid Withdrawal Decreases Balance
Class	BankAccountTest
Method	withdraw()
Precondition	Bank Account created with initial balance of \$100
Input	Amount = 60.0
Expected	Balance decreases to 40.0
Actual	Balance is 40.0
Status	PASS

Test ID	TC002
Test Name	Withdrawal more than balance throws exception
Class	BankAccountTest
Method	withdraw()
Precondition	Bank Account created with initial balance of \$100
Input	Amount = 101.0
Expected	Illegal state exception thrown - insufficient funds
Actual	Illegal state exception thrown - insufficient funds
Status	Passed

Test ID	TC003
Test Name	Withdrawal of Zero throws exception
Class	BankAccountTest
Method	withdraw()
Precondition	Bank Account created with initial balance of \$100
Input	Amount = 0.0
Expected	Illegal argument exception thrown - no amount withdrawn
Actual	Illegal argument exception thrown - no amount withdrawn
Status	PASS

Test ID	TC004
Test Name	Withdrawal of negative throws exception
Class	BankAccountTest
Method	withdraw()
Precondition	Bank Account created with initial balance of \$100
Input	Amount = -20.0
Expected	Illegal argument exception thrown - negative money can't be withdrawn
Actual	Illegal argument exception thrown - negative money can't be withdrawn
Status	PASS

Test ID	TC005
Test Name	Maximum Withdrawal Amount
Class	BankAccountTest
Method	withdraw()
Precondition	Bank Account created with initial balance of \$100
Input	Amount=100.0
Expected	Balance decreases to 0.0
Actual	Balance is 0.0
Status	PASS

Test ID	TC006
Test Name	Valid Deposit increases balance
Class	BankAccountTest
Method	deposit()
Precondition	Bank Account created with initial balance of \$100
Input	Amount = 50.0
Expected	Balance increases to 150.0
Actual	Balance is 150.0
Status	PASS

Test ID	TC007
Test Name	Deposit Negative Amount throws exception
Class	BankAccountTest
Method	deposit()
Precondition	Bank Account created with initial balance of \$100
Input	Amount = -20.0
Expected	Illegal argument exception thrown - negative money can't be deposited
Actual	Illegal argument exception thrown - negative money can't be deposited
Status	PASS

Test ID	TC008
Test Name	Deposit Zero Amount throws exception
Class	BankAccountTest
Method	deposit()
Precondition	Bank Account created with initial balance of \$100
Input	Amount = 0.0
Expected	Illegal argument exception thrown - zero deposit can't be made
Actual	Illegal argument exception thrown - zero deposit can't be made
Status	PASS

Test ID	TC009
Test Name	Bank account with negative Balance
Class	BankAccountTest
Method	BankAccount()
Precondition	BankAccount created with initial balance of \$-50.0
Input	Balance = -50.0
Expected	Illegal argument exception thrown - a new bank account can't be initiated with a negative balance
Actual	Illegal argument exception thrown - a new bank account can't be initiated with a negative balance
Status	PASS

Test ID	TC010
Test Name	Zero initial balance is valid
Class	BankAccountTest
Method	BankAccount()
Precondition	BankAccount created with initial balance of \$0.0
Input	Balance = 0.0
Expected	No increase/decrease in balance. Balance is 0.0
Actual	Balance is 0.0
Status	PASS

Notification Service Test Cases

Test ID	TC011
Test Name	processDeposit increases account balance
Class	TransactionServiceTest
Method	processDeposit()
Precondition	BankAccount created with initial balance of \$100.0 TransactionService created with mocked NotificationService
Input	Amount = 50.0
Expected	Amount increases to 150.0
Actual	Amountincreases to 150.0
Status	Pass

Test ID	TC012
Test Name	processDeposit sends notification alert
Class	TransactionServiceTest
Method	processDeposit()
Precondition	BankAccount created with initial balance of \$100.0 TransactionService created with mocked NotificationService
Input	Amount = 50.0
Expected	NotificationService.sendAlert() called with message "Deposit of 50.0 processed."
Actual	NotificationService.sendAlert() called with message "Deposit of 50.0 processed."
Status	PASS

Test ID	TC013
Test Name	processWithdrawal decreases account balance
Class	TransactionServiceTest
Method	processWithdrawal()
Precondition	BankAccount created with initial balance of \$100.0 TransactionService created with mocked NotificationService
Input	Amount = 30.0
Expected	Amount decreases to 70.0
Actual	Amount decreases to 70.0
Status	PASS

Test ID	TC014
Test Name	processWithdrawal sends notification alert
Class	TransactionServiceTest
Method	processWithdrawal()
Precondition	BankAccount created with initial balance of \$100.0 TransactionService created with mocked NotificationService
Input	Amount = 30.0
Expected	NotificationService.sendAlert() called with message "Withdrawal of 50.0 processed."
Actual	NotificationService.sendAlert() called with message "Withdrawal of 50.0 processed."
Status	PASS

Test ID	TC015
Test Name	processWithdrawal insufficient funds does not send notification
Class	TransactionServiceTest
Method	processWithdrawal
Precondition	BankAccount created with initial balance of \$100.0 TransactionService created with mocked NotificationService
Input	Amount = 500.0
Expected	Illegal State exception thrown - insufficient funds - no notification sent — verified using verifyNoInteractions(notificationService)
Actual	Illegal State exception thrown - insufficient funds - no notification sent — verified using verifyNoInteractions(notificationService)
Status	PASS

UserLogin Test

Test ID	TC016
Test Name	Valid Login Credentials return true
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: “User123” , password: “Password123”
Expected	login() returns true — valid credentials match
Actual	login() returns true — valid credentials match
Status	PASS

Test ID	TC017
Test Name	Wrong password return false
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: “User123” , password: “password12”
Expected	login() returns false — invalid password
Actual	login() returns false — invalid password
Status	PASS

Test ID	TC018
Test Name	Wrong username return false
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: “WrongUser” , password: “password12”
Expected	login() returns false — invalid username
Actual	login() returns false — invalid username
Status	PASS

Test ID	TC019
Test Name	Empty Username throws exception
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: "", password: "Password123"
Expected	Illegal Argument exception thrown - username can't be empty
Actual	Illegal Argument exception thrown - username can't be empty
Status	PASS

Test ID	TC019
Test Name	Empty Password throws exception
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: "User123", password: ""
Expected	login() returns false — invalid credentials
Actual	login() returns false — invalid credentials
Status	PASS

Test ID	TC021
Test Name	Password too short throws an exception
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: "User123", password: "pass"
Expected	Illegal Argument Exception thrown - password is too short
Actual	ERROR: IllegalArgumentException to be thrown, but nothing was thrown.
Status	FAIL - SEE BUG 001

Test ID	TC022
Test Name	Null Username throws exception
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: null , password: "Password123"
Expected	Illegal Argument Exception thrown - username can't be null
Actual	Illegal Argument Exception thrown - username can't be null
Status	PASS

Test ID	TC023
Test Name	Null Password throws exception
Class	UserLoginTest
Method	login()
Precondition	A new user is initiated with userName: User123 and password: Password123
Input	userName: "User123" , password: null
Expected	Illegal Argument Exception thrown - password can't be null
Actual	Illegal Argument Exception thrown - password can't be null
Status	PASS